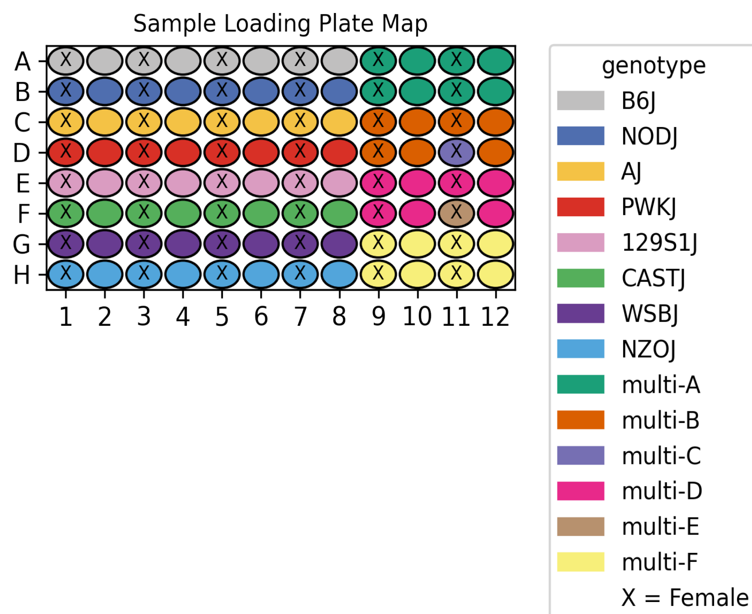
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_004
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	Heart Liver
Subpool	Subpool_7
Measurment Set	IGVFDS2344ZIPD
Exome capture subpool	No
Expected nuclei	67,000



1. Sample multiplexing

Heart was the main tissue on the plate while Liver was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	AJ
multi-D	129S1J	CASTJ
multi-E	CASTJ	PWKJ
multi-F	NZOJ	WSBJ

2. Alignment

A total of 1,497,664,817 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_7	1,497,664,817	767,959,322	0.256	1,031,757,413	68.9

3. Cell recovery

A total of 252,125 nuclei in Subpool_7 have >=200 UMIs and 81,861 nuclei have >= 500 UMIs.

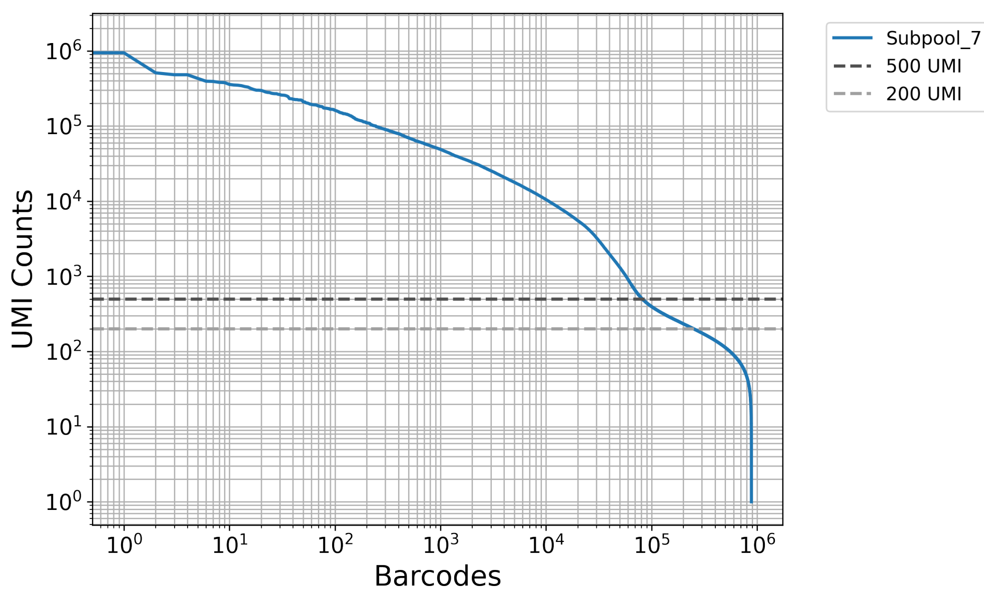


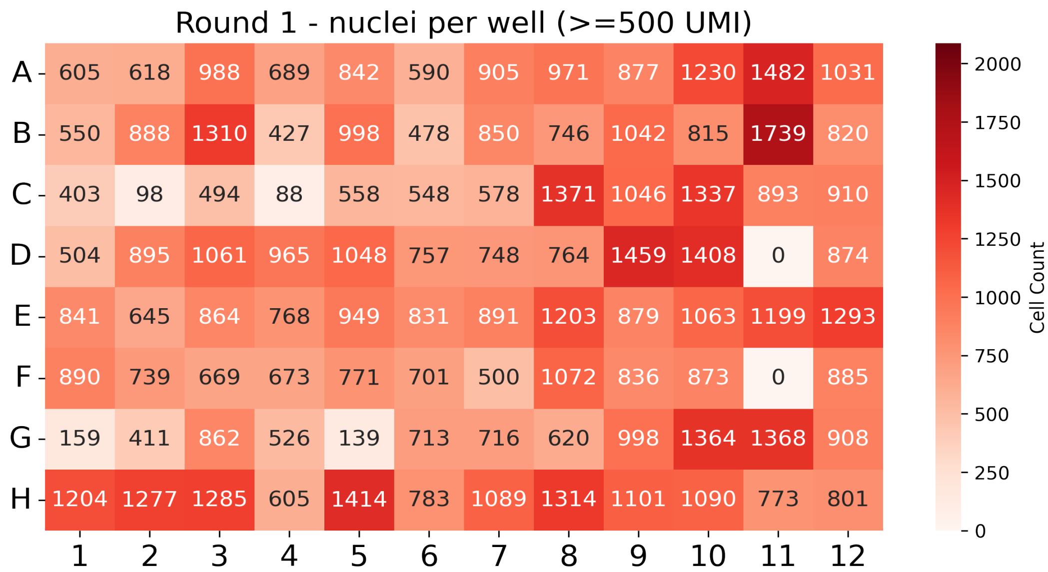
Table 3: QC stats for nuclei >=200 UMI

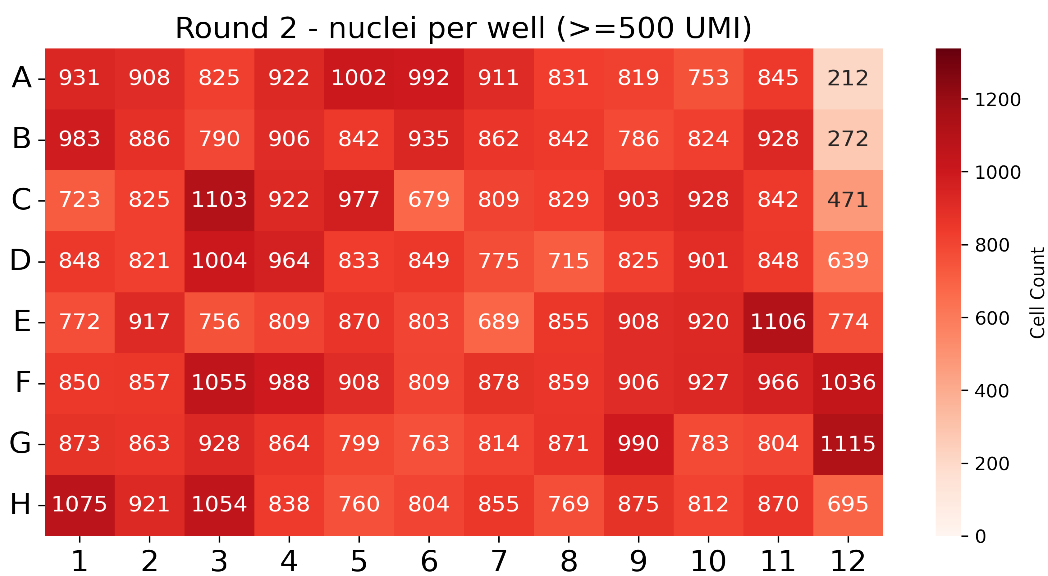
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_7	252,125	324	2,007.8	268	733.6

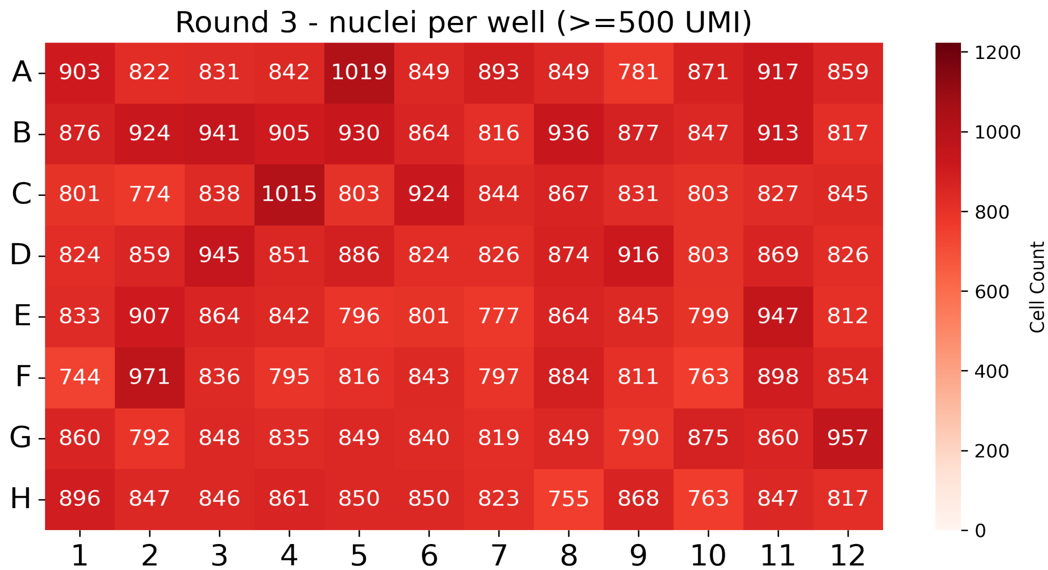
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_7	81,861	1,894	5,589.6	1,150	1,767.2

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







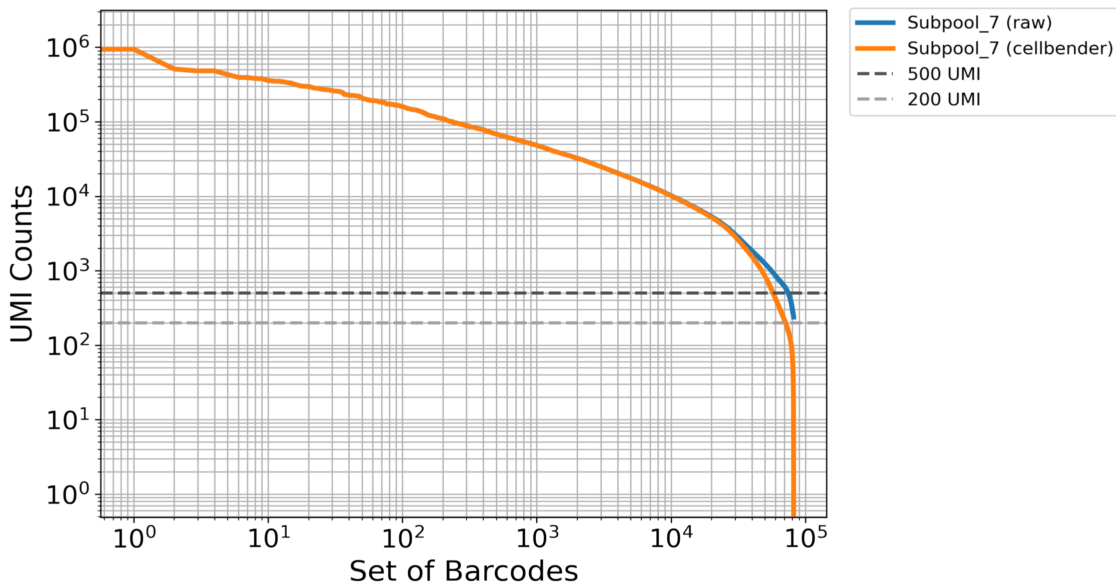
4. Background removal

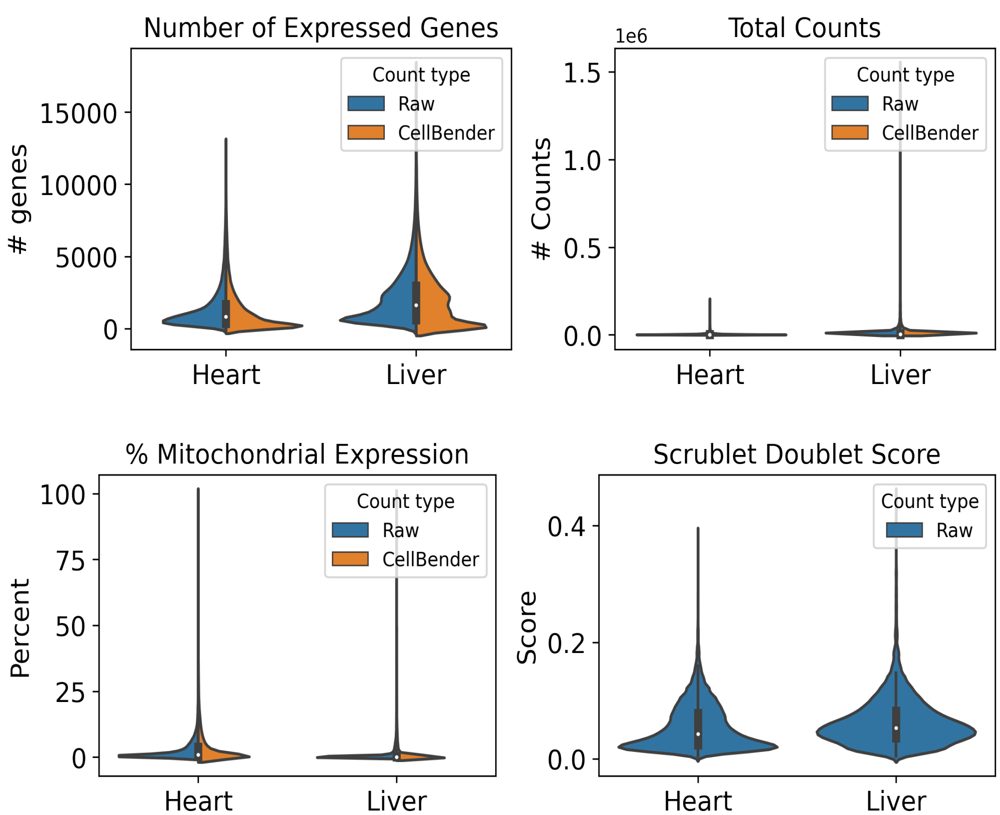
Table 5: Cellbender parameters

Subpool	droplets_included	learning_rate	expected_cells
Subpool_7	200000	5e-05	67000

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_7	5.0	83,671	5,190.9





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM2397OXJE	016_B6J_10F_06	16	R	A1
CATTCCTA	IGVFSM2397OXJE	016_B6J_10F_06	16	T	A1
CTGCTTTG	IGVFSM6821QIOI	017_B6J_10M_06	16	R	A2
CTTCATCA	IGVFSM6821QIOI	017_B6J_10M_06	16	T	A2
CTAAGGGA	IGVFSM9283STYP	018_B6J_10F_06	16	R	A3
CCTATATC	IGVFSM9283STYP	018_B6J_10F_06	16	T	A3
GCTTATAG	IGVFSM3819GVRA	019_B6J_10M_06	16	R	A4
ACATTTAC	IGVFSM3819GVRA	019_B6J_10M_06	16	T	A4
TCTGATCC	IGVFSM9188TGLO	020_B6J_10F_06	16	R	A5
ACTTAGCT	IGVFSM9188TGLO	020_B6J_10F_06	16	T	A5
TCTCTTGG	IGVFSM4152MQAV	021_B6J_10M_06	16	R	A6
CCAATTCT	IGVFSM4152MQAV	021_B6J_10M_06	16	T	A6
CAATTTCC	IGVFSM1715AZGK	024_B6J_10F_06	16	R	A7
GCCTATCT	IGVFSM1715AZGK	024_B6J_10F_06	16	T	A7
AGTCTCTT	IGVFSM0659JNLJ	025_B6J_10M_06	16	R	A8
ATGCTGCT	IGVFSM0659JNLJ	025_B6J_10M_06	16	T	A8
TGCTGCTC	IGVFSM2609IWQV	016_B6J_10F_05	16	R	A9
CATTTACA	IGVFSM2609IWQV	016_B6J_10F_05	16	T	A9
TGCTGCTC	IGVFSM7322GJHR	066_NODJ_10F_05	16	R	A9
CATTTACA	IGVFSM7322GJHR	066_NODJ_10F_05	16	T	A9
GTATTTCC	IGVFSM5795GRDC	017_B6J_10M_05	16	R	A10
ACTCGTAA	IGVFSM5795GRDC	017_B6J_10M_05	16	T	A10
GTATTTCC	IGVFSM1637CAZY	067_NODJ_10M_05	16	R	A10
ACTCGTAA	IGVFSM1637CAZY	067_NODJ_10M_05	16	T	A10
TTCCTGTG	IGVFSM9969QCIJ	018_B6J_10F_05	16	R	A11
CCTTTGCA	IGVFSM9969QCIJ	018_B6J_10F_05	16	T	A11
TTCCTGTG	IGVFSM8221YADP	068_NODJ_10F_05	16	R	A11

CCTTTGCA	IGVFSM8221YADP	068_NODJ_10F_05	16	T	A11
GCTGCTTC	IGVFSM6046EOEX	019_B6J_10M_05	16	R	A12
ACTCCTGC	IGVFSM6046EOEX	019_B6J_10M_05	16	T	A12
GCTGCTTC	IGVFSM2794WEMQ	069_NODJ_10M_05	16	R	A12
ACTCCTGC	IGVFSM2794WEMQ	069_NODJ_10M_05	16	T	A12
TATGTGTC	IGVFSM0400KSXT	066_NODJ_10F_06	16	R	B1
ATTTGGCA	IGVFSM0400KSXT	066_NODJ_10F_06	16	T	B1
CAATTCTC	IGVFSM8486IRSE	067_NODJ_10M_06	16	R	B2
TTATTCTG	IGVFSM8486IRSE	067_NODJ_10M_06	16	T	B2
TGGTCTCC	IGVFSM6061AIFT	068_NODJ_10F_06	16	R	B3
TCATGCTC	IGVFSM6061AIFT	068_NODJ_10F_06	16	T	B3
GCTCTTTA	IGVFSM9555SHCP	069_NODJ_10M_06	16	R	B4
CATACTTC	IGVFSM9555SHCP	069_NODJ_10M_06	16	T	B4
GCTGCATG	IGVFSM2548HETT	070_NODJ_10F_06	16	R	B5
CCGTTCTA	IGVFSM2548HETT	070_NODJ_10F_06	16	T	B5
ACTCATTT	IGVFSM6976MFYC	071_NODJ_10M_06	16	R	B6
GCTTCATA	IGVFSM6976MFYC	071_NODJ_10M_06	16	T	B6
AGTCTTGG	IGVFSM1795GGVQ	072_NODJ_10F_06	16	R	B7
CTCTGTGC	IGVFSM1795GGVQ	072_NODJ_10F_06	16	T	B7
GGTTCTTC	IGVFSM9138HBOW	073_NODJ_10M_06	16	R	B8
CCCTTATA	IGVFSM9138HBOW	073_NODJ_10M_06	16	T	B8
TCATGTTG	IGVFSM0660OYYA	020_B6J_10F_05	16	R	B9
ACTGCTCT	IGVFSM0660OYYA	020_B6J_10F_05	16	T	B9
TCATGTTG	IGVFSM1557PJAI	070_NODJ_10F_05	16	R	B9
ACTGCTCT	IGVFSM1557PJAI	070_NODJ_10F_05	16	T	B9
ATTTTGCC	IGVFSM9376MAOE	021_B6J_10M_05	16	R	B10
CTCTAATC	IGVFSM9376MAOE	021_B6J_10M_05	16	T	B10
ATTTTGCC	IGVFSM5227MWGM	071_NODJ_10M_05	16	R	B10
CTCTAATC	IGVFSM5227MWGM	071_NODJ_10M_05	16	T	B10
CTTCTGTA	IGVFSM4625KNRU	024_B6J_10F_05	16	R	B11
ACCCTTGC	IGVFSM4625KNRU	024_B6J_10F_05	16	T	B11
CTTCTGTA	IGVFSM8370FQBH	072_NODJ_10F_05	16	R	B11
ACCCTTGC	IGVFSM8370FQBH	072_NODJ_10F_05	16	T	B11
GTCCATCT	IGVFSM9141WALR	025_B6J_10M_05	16	R	B12
ATCTTAGG	IGVFSM9141WALR	025_B6J_10M_05	16	T	B12

GTCCATCT	IGVFSM3979KZZX	073_NODJ_10M_05	16	R	B12
ATCTTAGG	IGVFSM3979KZZX	073_NODJ_10M_05	16	T	B12
GCTATCTC	IGVFSM7523CIFZ	026_AJ_10F_06	16	R	C1
CATGTCTC	IGVFSM7523CIFZ	026_AJ_10F_06	16	T	C1
TAGTTTCC	IGVFSM5294ZSIL	029_AJ_10M_06	16	R	C2
TCATTGCA	IGVFSM5294ZSIL	029_AJ_10M_06	16	T	C2
TCCATTAT	IGVFSM1802ZPXC	028_AJ_10F_06	16	R	C3
ACACCTTT	IGVFSM1802ZPXC	028_AJ_10F_06	16	T	C3
AGGATTAA	IGVFSM4608HMDV	031_AJ_10M_06	16	R	C4
AATTTCTC	IGVFSM4608HMDV	031_AJ_10M_06	16	T	C4
AATCTTTC	IGVFSM5549SWHL	030_AJ_10F_06	16	R	C5
ATTCATGG	IGVFSM5549SWHL	030_AJ_10F_06	16	T	C5
GTCATATG	IGVFSM7128WVEJ	033_AJ_10M_06	16	R	C6
ACTTTACC	IGVFSM7128WVEJ	033_AJ_10M_06	16	T	C6
GTGCTTCC	IGVFSM2048BQHM	032_AJ_10F_06	16	R	C7
CTTCTAAC	IGVFSM2048BQHM	032_AJ_10F_06	16	T	C7
ATGTGTTG	IGVFSM2812EMHZ	035_AJ_10M_06	16	R	C8
CTATTTCA	IGVFSM2812EMHZ	035_AJ_10M_06	16	T	C8
CCATCTTG	IGVFSM7760MHIK	026_AJ_10F_05	16	R	C9
TCTCATGC	IGVFSM7760MHIK	026_AJ_10F_05	16	T	C9
CCATCTTG	IGVFSM8222BTFH	076_PWKJ_10F_05	16	R	C9
TCTCATGC	IGVFSM8222BTFH	076_PWKJ_10F_05	16	T	C9
TACTGTCT	IGVFSM8587LUPL	029_AJ_10M_05	16	R	C10
ATCCTTAC	IGVFSM8587LUPL	029_AJ_10M_05	16	T	C10
TACTGTCT	IGVFSM6195ZDYB	077_PWKJ_10M_05	16	R	C10
ATCCTTAC	IGVFSM6195ZDYB	077_PWKJ_10M_05	16	T	C10
TTCATCGC	IGVFSM8501AGYE	028_AJ_10F_05	16	R	C11
TAAATATC	IGVFSM8501AGYE	028_AJ_10F_05	16	T	C11
TTCATCGC	IGVFSM0598HJAL	078_PWKJ_10F_05	16	R	C11
TAAATATC	IGVFSM0598HJAL	078_PWKJ_10F_05	16	T	C11
ACTGTGGG	IGVFSM9889OUOI	031_AJ_10M_05	16	R	C12
TTACCTGC	IGVFSM9889OUOI	031_AJ_10M_05	16	T	C12
ACTGTGGG	IGVFSM7868PMHQ	079_PWKJ_10M_05	16	R	C12
TTACCTGC	IGVFSM7868PMHQ	079_PWKJ_10M_05	16	T	C12
TCTGTGCC	IGVFSM2489HRKN	076_PWKJ_10F_06	16	R	D1

CACTTTCA	IGVFSM2489HRKN	076_PWKJ_10F_06	16	T	D1
TCAATCTC	IGVFSM9967KFUW	077_PWKJ_10M_06	16	R	D2
CACCTTTA	IGVFSM9967KFUW	077_PWKJ_10M_06	16	T	D2
GTCCTCTG	IGVFSM9748KTTY	078_PWKJ_10F_06	16	R	D3
CTGACTTC	IGVFSM9748KTTY	078_PWKJ_10F_06	16	T	D3
TTACATTC	IGVFSM6943ARXK	079_PWKJ_10M_06	16	R	D4
CATTTGGA	IGVFSM6943ARXK	079_PWKJ_10M_06	16	T	D4
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GCTCTACT	IGVFSM8513LDPM	080_PWKJ_10F_06	16	T	D5
TGTGTATG	IGVFSM3449RSOM	081_PWKJ_10M_06	16	R	D6
GTTACGTA	IGVFSM3449RSOM	081_PWKJ_10M_06	16	T	D6
TCCATTTG	IGVFSM5687CRRX	084_PWKJ_10F_06	16	R	D7
CCTGTTGC	IGVFSM5687CRRX	084_PWKJ_10F_06	16	T	D7
TTAGCTTC	IGVFSM2657UOQA	085_PWKJ_10M_06	16	R	D8
CTATCATC	IGVFSM2657UOQA	085_PWKJ_10M_06	16	T	D8
GTGCTTGA	IGVFSM9122CTMU	030_AJ_10F_05	16	R	D9
GCTATCAT	IGVFSM9122CTMU	030_AJ_10F_05	16	T	D9
GTGCTTGA	IGVFSM4205CGEZ	080_PWKJ_10F_05	16	R	D9
GCTATCAT	IGVFSM4205CGEZ	080_PWKJ_10F_05	16	T	D9
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GTTTGTGA	IGVFSM9316HSZQ	081_PWKJ_10M_05	16	R	D10
ACATTCAT	IGVFSM9316HSZQ	081_PWKJ_10M_05	16	T	D10
GAAATTAG	IGVFSM3749AQYK	032_AJ_10F_05	16	R	D11
TTCGCTAC	IGVFSM3749AQYK	032_AJ_10F_05	16	T	D11
GAAATTAG	IGVFSM2116DCHR	044_129S1J_10F_05	16	R	D11
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TTTGCATC	IGVFSM9345WJHN	041_129S1J_10M_06	16	R	E4
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AGGTAATA	IGVFSM5009WOZO	040_129S1J_10F_06	16	R	E5
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GTGCCTTC	IGVFSM5004FHRH	043_129S1J_10M_06	16	R	E6
TTGGGAGA	IGVFSM5004FHRH	043_129S1J_10M_06	16	T	E6
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CTTAATTC	IGVFSM5577TDWZ	045_129S1J_10M_06	16	R	E8
TCTCTAGC	IGVFSM5577TDWZ	045_129S1J_10M_06	16	T	E8
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TCTGGCTC	IGVFSM7228AIWD	086_CASTJ_10F_05	16	R	E9
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CATCATTT	IGVFSM3344MCNS	037_129S1J_10M_05	16	R	E10
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CATCATTT	IGVFSM9408IVIR	087_CASTJ_10M_05	16	R	E10
CCCAATTT	IGVFSM9408IVIR	087_CASTJ_10M_05	16	T	E10
GTTGTCTC	IGVFSM9049MUBB	038_129S1J_10F_05	16	R	E11
ACTATATA	IGVFSM9049MUBB	038_129S1J_10F_05	16	T	E11
GTTGTCTC	IGVFSM3996IQCS	088_CASTJ_10F_05	16	R	E11
ACTATATA	IGVFSM3996IQCS	088_CASTJ_10F_05	16	T	E11
ATCTTCTG	IGVFSM6336XNLQ	041_129S1J_10M_05	16	R	E12
CTCTATAC	IGVFSM6336XNLQ	041_129S1J_10M_05	16	T	E12
ATCTTCTG	IGVFSM0151EMWS	089_CASTJ_10M_05	16	R	E12
CTCTATAC	IGVFSM0151EMWS	089_CASTJ_10M_05	16	T	E12
TGTTTGCC	IGVFSM5819SOLY	086_CASTJ_10F_06	16	R	F1
CTGTCTCA	IGVFSM5819SOLY	086_CASTJ_10F_06	16	T	F1
TTCTGTCA	IGVFSM0487BHBU	087_CASTJ_10M_06	16	R	F2
GACCTTTC	IGVFSM0487BHBU	087_CASTJ_10M_06	16	T	F2
ACGGACTC	IGVFSM5847JXKA	088_CASTJ_10F_06	16	R	F3
GATTTGGC	IGVFSM5847JXKA	088_CASTJ_10F_06	16	T	F3
TTTGGTCA	IGVFSM4021BOPQ	089_CASTJ_10M_06	16	R	F4

CGTCTAGG	IGVFSM4021BOPQ	089_CASTJ_10M_06	16	T	F4
TATCCGGG	IGVFSM4983NNIU	090_CASTJ_10F_06	16	R	F5
TACTCGAA	IGVFSM4983NNIU	090_CASTJ_10F_06	16	T	F5
TGTCATTC	IGVFSM2511KNVM	091_CASTJ_10M_06	16	R	F6
CAGCCTTT	IGVFSM2511KNVM	091_CASTJ_10M_06	16	T	F6
ATTCTCTG	IGVFSM8621PXJN	094_CASTJ_10F_06	16	R	F7
CCTCATTA	IGVFSM8621PXJN	094_CASTJ_10F_06	16	T	F7
TGGCTTCC	IGVFSM0998PUXL	093_CASTJ_10M_06	16	R	F8
CTTATACC	IGVFSM0998PUXL	093_CASTJ_10M_06	16	T	F8
TTGTTGCC	IGVFSM7035ACUG	040_129S1J_10F_05	16	R	F9
TCTATTAC	IGVFSM7035ACUG	040_129S1J_10F_05	16	T	F9
TTGTTGCC	IGVFSM6880XYMR	090_CASTJ_10F_05	16	R	F9
TCTATTAC	IGVFSM6880XYMR	090_CASTJ_10F_05	16	T	F9
GTCATCTC	IGVFSM5467HYTI	043_129S1J_10M_05	16	R	F10
CCTGCATT	IGVFSM5467HYTI	043_129S1J_10M_05	16	T	F10
GTCATCTC	IGVFSM9848ACFP	091_CASTJ_10M_05	16	R	F10
CCTGCATT	IGVFSM9848ACFP	091_CASTJ_10M_05	16	T	F10
TTGCTCAT	IGVFSM3421MHKE	084_PWKJ_10F_05	16	R	F11
CAATCCTT	IGVFSM3421MHKE	084_PWKJ_10F_05	16	T	F11
TTGCTCAT	IGVFSM1855TZJL	094_CASTJ_10F_05	16	R	F11
CAATCCTT	IGVFSM1855TZJL	094_CASTJ_10F_05	16	T	F11
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TTGTCTTA	IGVFSM9876UKZX	045_129S1J_10M_05	16	T	F12
CTGTCTGC	IGVFSM8245FNTV	093_CASTJ_10M_05	16	R	F12
TTGTCTTA	IGVFSM8245FNTV	093_CASTJ_10M_05	16	T	F12
TATATTCC	IGVFSM3432ZCOX	058_WSBJ_10F_06	16	R	G1
TCACTTTA	IGVFSM3432ZCOX	058_WSBJ_10F_06	16	T	G1
ATATTGGC	IGVFSM1456DXFQ	057_WSBJ_10M_06	16	R	G2
TGCTTGGG	IGVFSM1456DXFQ	057_WSBJ_10M_06	16	T	G2
GTGTCCTC	IGVFSM7766OOSU	060_WSBJ_10F_06	16	R	G3
CGCTCATT	IGVFSM7766OOSU	060_WSBJ_10F_06	16	T	G3
ATCTTCAT	IGVFSM6894ALPA	059_WSBJ_10M_06	16	R	G4
GCCTCTAT	IGVFSM6894ALPA	059_WSBJ_10M_06	16	T	G4
CGTGGTTG	IGVFSM3113AHHA	062_WSBJ_10F_06	16	R	G5
GAGCACAA	IGVFSM3113AHHA	062_WSBJ_10F_06	16	T	G5

TTGCATCC	IGVFSM8737KUQW	061_WSBJ_10M_06	16	R	G6
CTCTTAAC	IGVFSM8737KUQW	061_WSBJ_10M_06	16	T	G6
TCTTAATC	IGVFSM6802LFPK	096_WSBJ_10F_06	16	R	G7
TCTAGGCT	IGVFSM6802LFPK	096_WSBJ_10F_06	16	T	G7
TGCATTTC	IGVFSM6965RRET	063_WSBJ_10M_06	16	R	G8
AATTCTGC	IGVFSM6965RRET	063_WSBJ_10M_06	16	T	G8
GATGTTTC	IGVFSM9090ZNRS	046_NZOJ_10F_05	16	R	G9
CATTCTCA	IGVFSM9090ZNRS	046_NZOJ_10F_05	16	T	G9
GATGTTTC	IGVFSM6714CAWV	058_WSBJ_10F_05	16	R	G9
CATTCTCA	IGVFSM6714CAWV	058_WSBJ_10F_05	16	T	G9
ATCTTGTC	IGVFSM4130EMOU	047_NZOJ_10M_05	16	R	G10
ACTTGCCT	IGVFSM4130EMOU	047_NZOJ_10M_05	16	T	G10
ATCTTGTC	IGVFSM6424UEBK	057_WSBJ_10M_05	16	R	G10
ACTTGCCT	IGVFSM6424UEBK	057_WSBJ_10M_05	16	T	G10
TCATATTC	IGVFSM1384SGZS	048_NZOJ_10F_05	16	R	G11
ATCATTGC	IGVFSM1384SGZS	048_NZOJ_10F_05	16	T	G11
TCATATTC	IGVFSM0803WGCL	060_WSBJ_10F_05	16	R	G11
ATCATTGC	IGVFSM0803WGCL	060_WSBJ_10F_05	16	T	G11
TGGCCTCT	IGVFSM6644MYFA	049_NZOJ_10M_05	16	R	G12
GTTCAACA	IGVFSM6644MYFA	049_NZOJ_10M_05	16	T	G12
TGGCCTCT	IGVFSM0434YXCE	059_WSBJ_10M_05	16	R	G12
GTTCAACA	IGVFSM0434YXCE	059_WSBJ_10M_05	16	T	G12
CGTTGTCT	IGVFSM0510YNAJ	046_NZOJ_10F_06	16	R	H1
CCATTTGC	IGVFSM0510YNAJ	046_NZOJ_10F_06	16	T	H1
TCTTGTCA	IGVFSM7577ZSZL	047_NZOJ_10M_06	16	R	H2
GACTTTGC	IGVFSM7577ZSZL	047_NZOJ_10M_06	16	T	H2
TATTCCTG	IGVFSM4204PXKJ	048_NZOJ_10F_06	16	R	H3
ATTGGCTC	IGVFSM4204PXKJ	048_NZOJ_10F_06	16	T	H3
TCCATGTC	IGVFSM8056EVEE	049_NZOJ_10M_06	16	R	H4
GTGCTAGC	IGVFSM8056EVEE	049_NZOJ_10M_06	16	T	H4
TTGTCATC	IGVFSM8878JBBO	050_NZOJ_10F_06	16	R	H5
CTTTCAAC	IGVFSM8878JBBO	050_NZOJ_10F_06	16	T	H5
ATTTCCTG	IGVFSM2214BDBR	051_NZOJ_10M_06	16	R	H6
ACTATTGC	IGVFSM2214BDBR	051_NZOJ_10M_06	16	T	H6
GTGTCTCC	IGVFSM8146OBNU	052_NZOJ_10F_06	16	R	H7

ACTGGCTT	IGVFSM8146OBNU	052_NZOJ_10F_06	16	T	H7
GTGTGTGT	IGVFSM4952NHSA	053_NZOJ_10M_06	16	R	H8
ATTAGGCT	IGVFSM4952NHSA	053_NZOJ_10M_06	16	T	H8
TATGCTTC	IGVFSM9429JZJC	050_NZOJ_10F_05	16	R	H9
GCCTTTCA	IGVFSM9429JZJC	050_NZOJ_10F_05	16	T	H9
TATGCTTC	IGVFSM1705RPMT	062_WSBJ_10F_05	16	R	H9
GCCTTTCA	IGVFSM1705RPMT	062_WSBJ_10F_05	16	T	H9
ATGGTGTT	IGVFSM1239ETDH	051_NZOJ_10M_05	16	R	H10
ATTCTAGG	IGVFSM1239ETDH	051_NZOJ_10M_05	16	T	H10
ATGGTGTT	IGVFSM4873REQN	061_WSBJ_10M_05	16	R	H10
ATTCTAGG	IGVFSM4873REQN	061_WSBJ_10M_05	16	T	H10
GAATAATG	IGVFSM3658ESYZ	052_NZOJ_10F_05	16	R	H11
CCTTACAT	IGVFSM3658ESYZ	052_NZOJ_10F_05	16	T	H11
GAATAATG	IGVFSM3444WPKV	096_WSBJ_10F_05	16	R	H11
CCTTACAT	IGVFSM3444WPKV	096_WSBJ_10F_05	16	T	H11
CCTCTGTG	IGVFSM7821JHFX	053_NZOJ_10M_05	16	R	H12
ACATTTGG	IGVFSM7821JHFX	053_NZOJ_10M_05	16	T	H12
CCTCTGTG	IGVFSM3622HQCN	063_WSBJ_10M_05	16	R	H12
ACATTTGG	IGVFSM3622HQCN	063_WSBJ_10M_05	16	T	H12

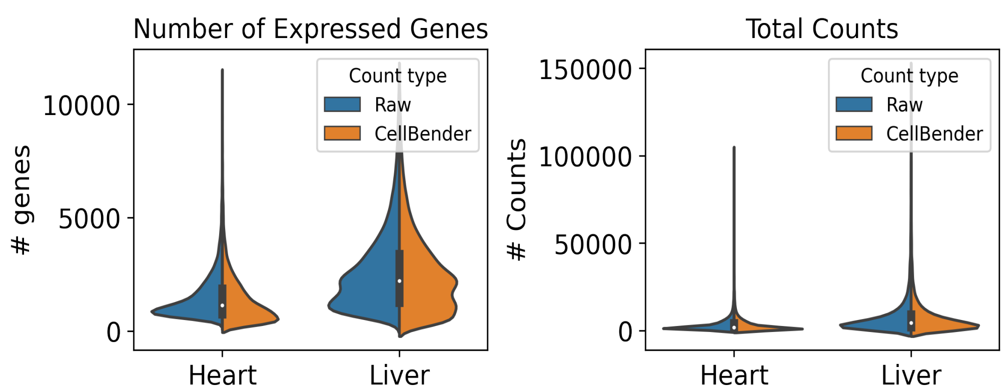
6. Description of data processing workflow and h5ad files

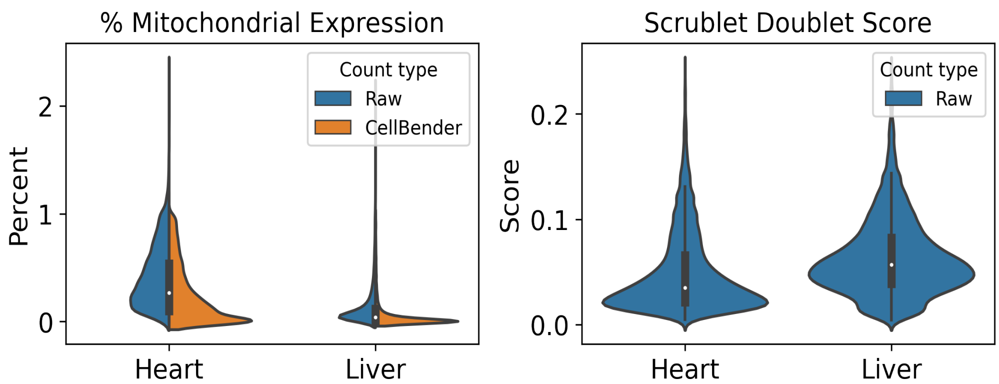
Data from Subpool_7 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

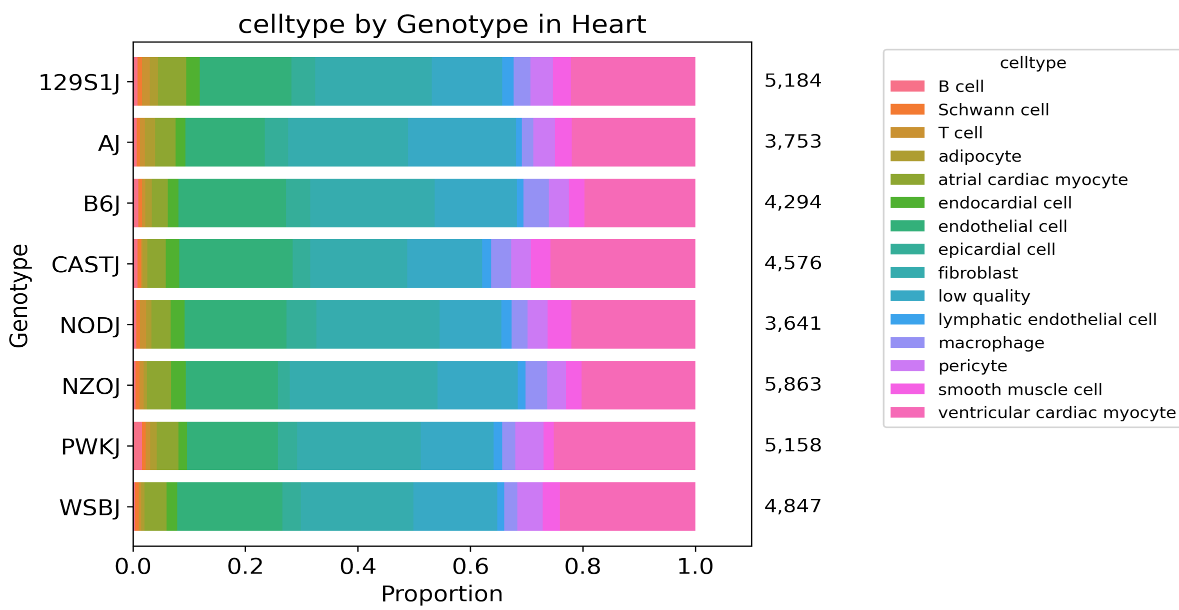
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 33 total clusters in Heart and 53 clusters in Liver, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



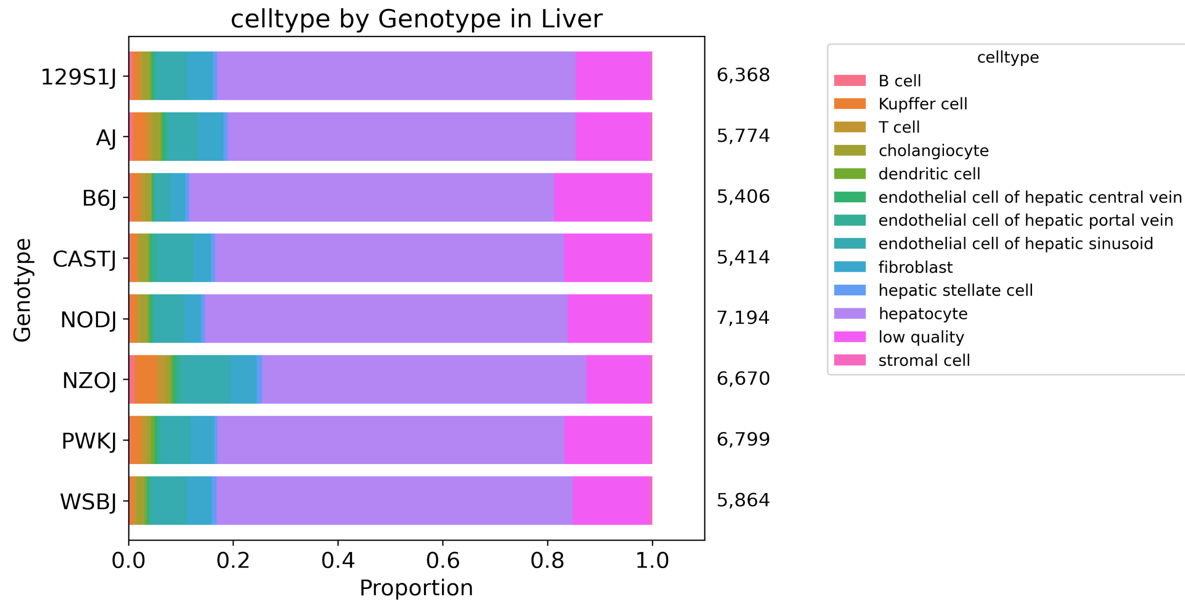


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_004.ipynb

Code used to process and annotate the data for main tissue:
https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Heart.ipynb

Code used to process and annotate the data for multiplexed tissue:
https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Liver.ipynb